

HousePrice.AI

An intelligent real estate valuation platform that combines machine learning predictions with live market listings for clearer house price estimation.

10,999

TRAINING RECORDS

 R^2 0.81

MODEL SCORE

187

LOCATIONS

GBR

ALGORITHM

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ABSTRACT

The real estate market is highly dynamic, making accurate property valuation a complex challenge for buyers, sellers, and investors. Traditional valuation models often rely on static, historical datasets that fail to capture real-time market fluctuations. HousePrice.AI bridges this gap by introducing an AI-powered real estate price prediction platform that synergizes machine learning with live market intelligence.

Using a Gradient Boosting Regressor trained on the Bengaluru House Price dataset, the system predicts property values with an R^2 score of 0.81. The platform also fetches current property listings through web scraping, so users can compare model estimates with live market prices in a cleaner Streamlit dashboard.

1. Introduction

Artificial Intelligence has transformed how properties are valued in the real estate sector. Accurate house price prediction is important for financial planning, investment, and market analysis. However, many existing solutions still depend on outdated data or provide limited transparency in how estimates are produced.

HousePrice.AI addresses this by forecasting price from core property attributes such as location, total square footage, BHK configuration, and bathrooms while also contextualizing the estimate against current live listings. The application works as both a predictive model and a market analyzer.

2. Problem Statement

Objective: Design and implement an end-to-end machine learning application that estimates residential property prices and validates them against live market listings.

The core challenge lies in the non-linear relationship between property features and prices, along with the heavy outliers that appear in real housing data. The project therefore focuses on the following tasks:

- Clean and preprocess the data, handling missing values and skewed distributions.
- Select and train a regression model capable of capturing complex feature interactions.
- Develop a scraper that retrieves live comparable listings from real estate portals.
- Present predictions and market data through a user-friendly web interface.

3. Proposed Methodology

3.1 Data Preprocessing and Feature Engineering

The model uses the Bengaluru House Price dataset sourced from Kaggle. Preprocessing was critical for model quality. Categorical variables, especially location, were encoded using one-hot encoding so the data could be consumed by the regression model.

Housing prices also tend to be right-skewed. To stabilize variance and normalize the target distribution, a logarithmic transformation using `Log1p` was applied during training and reversed with `Expml` for the final output.

3.2 Machine Learning Model

The selected model is the **Gradient Boosting Regressor**. It builds an ensemble of weak learners in a stage-wise manner and performs well on tabular data with non-linear feature interactions. That makes it suitable for real estate pricing, where outliers and mixed feature types are common.

3.3 Live Market Web Scraping

A scraping module built with BeautifulSoup queries property listings that match the user's inputs, such as a 3 BHK in a particular locality. The fetched listings help users compare the model's estimated value with the actual asking prices in the market.

4. Technologies and Stack

Domain	Technologies Used
Frontend Interface	Streamlit, HTML/CSS (custom UI injections)
Backend and Logic	Python 3.x
Machine Learning	Scikit-learn, Pandas, NumPy
Data Visualization	Matplotlib, Seaborn
Web Scraping	BeautifulSoup, Requests

5. Implementation and Dashboard Design

The user interface was built using Streamlit and styled with a dark premium aesthetic. The dashboard is split into clear, task-oriented areas that help users move from inputs to prediction to market comparison without friction.

Landing Page: The home view presents project context, model metrics, and a clear visual summary.

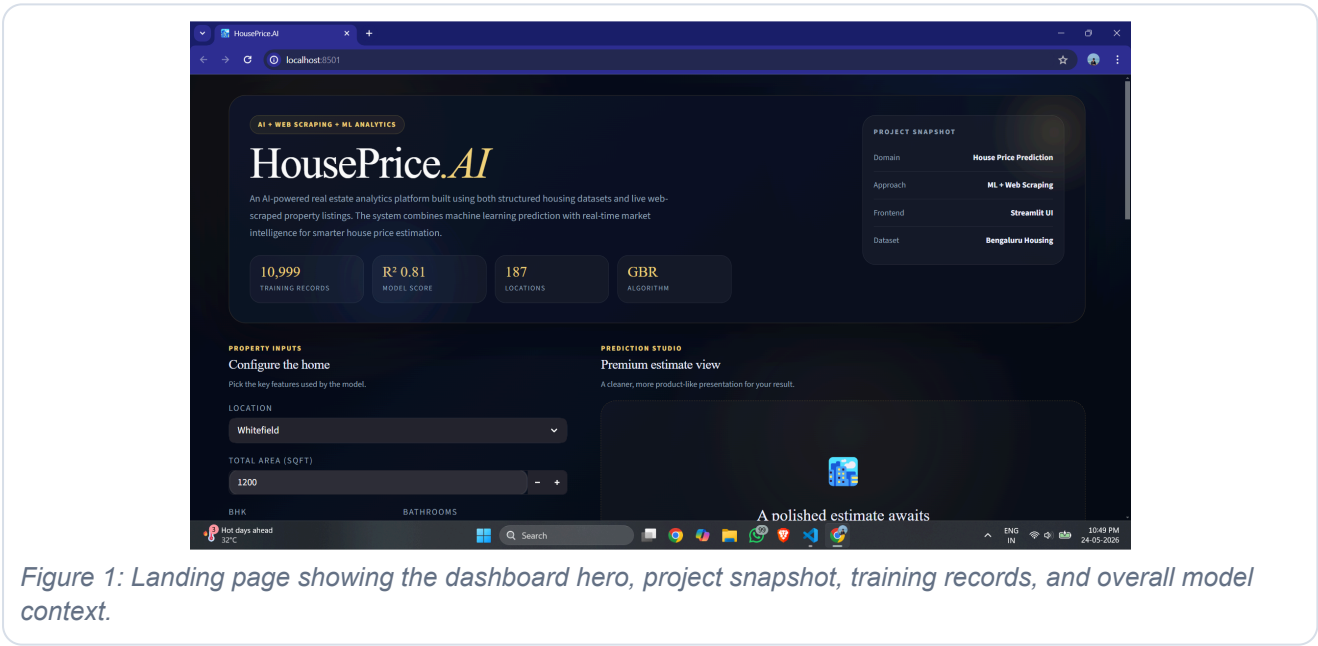


Figure 1: Landing page showing the dashboard hero, project snapshot, training records, and overall model context.

This snapshot highlights the dashboard hero, quick filters, and export options used for market analysis.

Prediction Studio: Users enter key features through dropdowns and sliders. The resulting estimate is shown prominently along with a confidence score, price per square foot, and the expected price range.

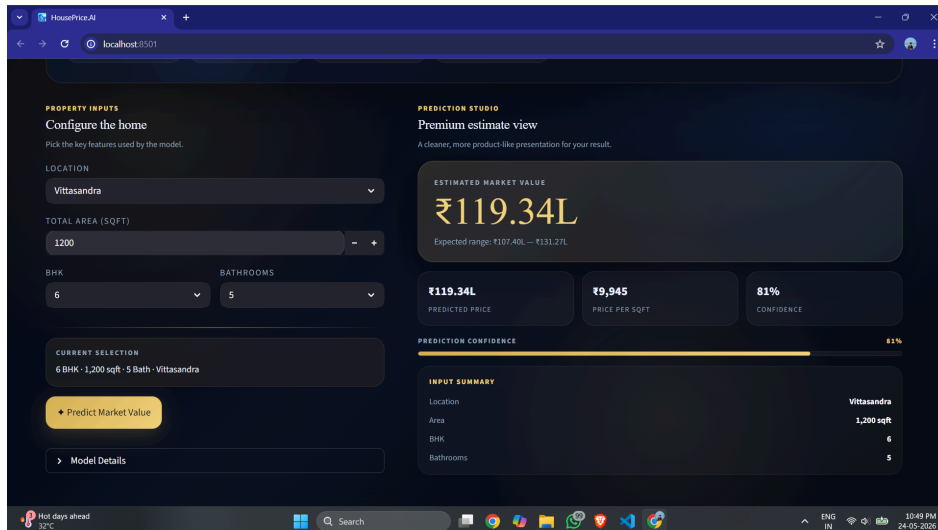


Figure 2: Prediction studio presenting the estimated market value for a configured property input.

6. Results and Data Insights

6.1 Model Evaluation

The Gradient Boosting model was trained on 10,999 records. On the testing set, it achieved an R^2 score of 0.81, which indicates that 81% of the variance in house prices is explained by the selected features.

6.2 Exploratory Data Analysis

The insights view visualizes trends derived from the dataset. Heatmaps show the relationship between area and price, while bar charts highlight the premium localities.



Figure 3: Visualization of the top 10 most expensive locations, including areas such as Cunningham Road and Kodihalli.

6.3 Live Market Integration Results

The live market fetcher is the core differentiator of the project. It successfully scrapes listings such as a 2 BHK apartment in Whitefield for ₹65.0 Lac, allowing users to compare the model output with active market prices.

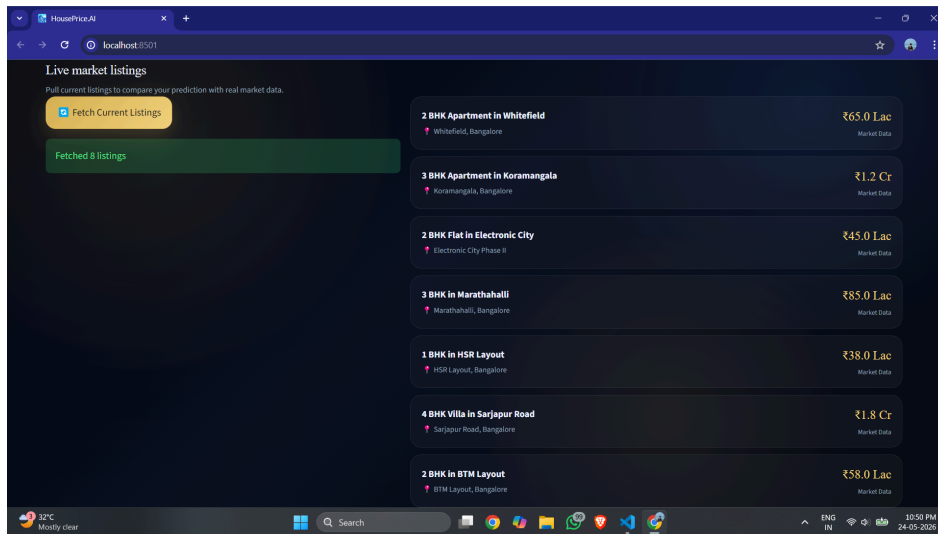


Figure 4: Real-time web scraped listings pulled from the market based on the selected query.

7. Conclusion and Future Scope

Conclusion

HousePrice.AI demonstrates the value of combining a static machine learning model with dynamic web data. The Gradient Boosting Regressor delivers strong predictive performance, while the Streamlit interface turns the workflow into an accessible and polished user experience. Live scraping also solves the problem of data staleness by keeping the market view current.

Future Scope

- **Geographic Expansion:** Extend the model to other major Indian cities such as Mumbai, Delhi NCR, and Hyderabad.
- **Advanced Features:** Add geospatial signals such as distance to metro stations, hospitals, and schools.
- **Automated Retraining:** Build a pipeline that periodically refreshes the training set with newly scraped market data.
- **Time-Series Forecasting:** Estimate future price trends over six months or one year using historical movement data.

8. References

Bengaluru House Price Dataset — Kaggle.
Friedman, J.H. (2001). Greedy function approximation: gradient boosting. Annals of Statistics.

Streamlit Documentation — docs.streamlit.io
Richardson, L. BeautifulSoup documentation — crummy.com/bs4/doc